

## PCI RINGS AND V-DOMAINS ARE NOT LEFT-RIGHT SYMMETRIC

**ABSTRACT.** We construct a countable domain  $R$  which is a left PCI domain—every proper cyclic left  $R$ -module is injective—but which is not right Ore. Consequently  $R$  is neither a right V-ring nor a right PCI ring, while its opposite ring  $R^{\text{op}}$  is a right PCI domain which is not left PCI, not a left V-ring, and not left Ore. This answers, in the negative, questions of Cozzens and Faith, of Damiano, and of Jain, Lam and Leroy. The ring is an Ore extension  $K[t; \sigma, \delta]$  in which  $K$  is a free field in the sense of Cohn and  $\sigma$  is a non-surjective endomorphism. The endomorphism and the derivation are prescribed on the free generators of  $K$  in such a way that every linear equation governing the divisibility of a cyclic module  $R/Rf$  is solved by a vector of free generators.

### 1. INTRODUCTION

A ring  $R$  is a *right PCI ring* if every proper cyclic right  $R$ -module—that is, every cyclic right module not isomorphic to  $R_R$ —is injective, and a *right V-ring* if every simple right  $R$ -module is injective; left PCI rings and left V-rings are defined symmetrically. Faith [Fai73] showed that a right PCI ring is either semisimple Artinian or a simple right semihereditary right Ore domain, and Damiano [Dam79, Theorem 1] showed that in the second case it is right Noetherian. Examples of PCI domains (on both sides) were given by Cozzens [Coz70]. Whether every right PCI ring is left PCI has been open since [Fai73]; see [BG75, §2], [CF75, §7] and [JS09, §2]. Cozzens and Faith [CF75, 6.25, 6.26], using a theorem of Boyle [Boy74], showed that a right Noetherian right PCI domain which is left Ore is left PCI; since right Noetherianity is automatic by Damiano’s theorem, the question is whether a right PCI domain must be left Ore. The companion question, whether a left V-domain must be a right V-domain, goes back to Cozzens and Faith [CF75, §7, (10)] and is restated as open in [JLL09, §1] and [BDV18, Remark 2.8]. Cozzens and Faith [CF75, §7, (7), (10)] proposed to look for counterexamples among Ore extensions  $K[t; \sigma, \delta]$  over a division ring  $K$  with a non-surjective endomorphism  $\sigma$ . Jain, Lam and Leroy [JLL09, §6] took this up: they showed that a left V-domain of this form is right PCI if and only if  $\sigma$  is onto [JLL09, Theorem 6.2], and remarked that they knew of no left V-domain  $K[t; \sigma, \delta]$  with  $\sigma$  not onto [JLL09, Remark 6.3].

We produce such an example.

**Theorem 1.1** (Main Theorem). *Let  $k$  be a countable commutative field. There are a countable division ring  $K \supseteq k$ , a non-surjective endomorphism  $\sigma$  of  $K$  and a  $\sigma$ -derivation  $\delta$  of  $K$  such that the Ore extension  $R = K[t; \sigma, \delta]$  has the following properties.*

- (1)  *$R$  is a simple principal left ideal domain and a left PCI domain; in particular  $R$  is a left V-domain.*
- (2)  *$R$  is not right Ore, not a right PCI ring, and not a right V-ring.*

*Consequently  $R^{\text{op}}$  is a simple right Noetherian right hereditary right PCI domain which is not left Ore, not a left V-ring, and not left PCI.*

The proof has two independent halves.

The first half (Section 2) is elementary ring theory over an arbitrary division ring  $K$  with an endomorphism  $\sigma$  and a  $\sigma$ -derivation  $\delta$ . If  $f \in R = K[t; \sigma, \delta]$  is monic of degree  $n$ , the cyclic module  $R/Rf$  is the row space  $K^n$  with  $t$  acting by a pseudo-linear map  $T_f$ , and  $R/Rf$

is injective as soon as the operators  $g(T_f)$  on  $K^n$ , for non-zero  $g \in R$ , are all surjective; this is Baer's criterion, and it gives a criterion for  $R$  to be left PCI (Proposition 2.2). If  $\sigma$  is not surjective then  $R$  is not right Ore (Proposition 2.3), and a cardinality argument shows that a countable such  $R$  has no non-zero countable injective right module, so is neither a right V-ring nor a right PCI ring (Lemma 2.4). The Main Theorem is thereby reduced to Theorem 2.5: there is a countable  $(K, \sigma, \delta)$  with  $\sigma$  not surjective over which all the operators  $g(T_f)$  are surjective.

The second half (Sections 3 and 4) constructs  $(K, \sigma, \delta)$ . We take  $K$  to be Cohn's free field on a countable set  $Y$  of generators. The only input from Cohn's theory is that the free algebra  $k\langle Y \rangle$  has a universal field of fractions; from this we derive, following Cohn, that an endomorphism and a twisted derivation of  $K$  may be prescribed arbitrarily on  $Y$  (Proposition 3.5), and that the endomorphism is not surjective if its restriction to  $Y$  is not (Lemma 3.6). The construction itself rests on a single observation. If  $x \in K^n$  is a vector whose coordinates are fixed by  $\sigma$ , then  $T_f(x) = xC_f + \delta(x)$ , where  $C_f$  is the companion matrix of  $f$ ; so to prescribe  $\delta$  on the coordinates of  $x$  is the same as to prescribe  $T_f(x)$ . For each equation  $g(T_f)(v) = e$ , with  $f$  and  $g$  monic of degrees  $n$  and  $m$ , we therefore reserve  $nm$  generators fixed by  $\sigma$ , arrange them into vectors  $x_0, \dots, x_{m-1} \in K^n$ , and decree that  $T_f(x_j) = x_{j+1}$  for  $j < m-1$  and  $T_f(x_{m-1}) = e - \sum_{j < m} g_j x_j$ ; then  $g(T_f)(x_0) = e$ . These prescriptions involve only the coefficients of  $f$ ,  $g$  and  $e$ , and not  $\sigma$  or  $\delta$  themselves, so all of them can be imposed at once. No tower of fields and no compatibility conditions are needed.

**Conventions.** Rings have an identity, and homomorphisms preserve it;  $k$  is a commutative field. For a division ring  $K$ , an *endomorphism*  $\sigma$  of  $K$  is a ring endomorphism (automatically injective), and a  $\sigma$ -*derivation* of  $K$  is an additive map  $\delta: K \rightarrow K$  with

$$\delta(ab) = \sigma(a)\delta(b) + \delta(a)b \quad (a, b \in K).$$

A *division subring* of a division ring is a subring closed under inverses of non-zero elements. For a map  $\alpha$  defined on  $K$  and a row vector or matrix  $v$  over  $K$ , we write  $\alpha(v)$  for the entrywise image. Compositions of maps are written by juxtaposition:  $\phi\psi = \phi \circ \psi$ .

## 2. ORE EXTENSIONS OVER DIVISION RINGS

Throughout this section  $K$  is a division ring with an endomorphism  $\sigma$  and a  $\sigma$ -derivation  $\delta$ . The *Ore extension*  $R = K[t; \sigma, \delta]$  is the ring which is a free left  $K$ -module with basis  $1, t, t^2, \dots$  and whose multiplication is determined by

$$ta = \sigma(a)t + \delta(a) \quad (a \in K).$$

Its elements are written  $f = \sum_i f_i t^i$  with the coefficients on the left, and  $\deg f$  is the usual degree. Since  $\sigma$  is injective,  $\deg(fg) = \deg f + \deg g$ ; so  $R$  is a domain, and its units are the non-zero elements of  $K$ . There is a left division algorithm: for  $f \neq 0$ , every  $r \in R$  can be written uniquely as  $r = qf + b$  with  $\deg b < \deg f$  (compare [CohFIR, Theorem 1.3.2], where the coefficients are written on the right). Hence every left ideal of  $R$  is of the form  $Rf$ , and  $R$  is a *principal left ideal domain*; in particular  $R$  is left Noetherian and left hereditary. Since  $R$  is a domain,  $Rf \cong R$  as left  $R$ -modules whenever  $f \neq 0$ .

**2.1. The modules  $R/Rf$ .** Let  $f = t^n + f_{n-1}t^{n-1} + \dots + f_0 \in R$  be monic of degree  $n \geq 1$ . By the division algorithm, the classes of  $1, t, \dots, t^{n-1}$  form a basis of  $R/Rf$  as a left  $K$ -vector space, and we identify  $R/Rf$  with the space  $K^n$  of row vectors via

$$\sum_{i < n} b_i t^i + Rf \longmapsto (b_0, \dots, b_{n-1}).$$

Let  $C_f \in M_n(K)$  be the companion matrix of  $f$ , whose rows are  $e_2, \dots, e_n, (-f_0, \dots, -f_{n-1})$ , where  $e_1, \dots, e_n$  are the standard basis vectors. Define

$$T_f: K^n \rightarrow K^n, \quad T_f(v) = \sigma(v)C_f + \delta(v).$$

The map  $T_f$  is additive and *pseudo-linear*:  $T_f(av) = \sigma(a)T_f(v) + \delta(a)v$  for  $a \in K$ . For  $g = \sum_{i=0}^m g_i t^i \in R$  we put

$$g(T_f)(v) = \sum_{i=0}^m g_i T_f^i(v) \quad (v \in K^n),$$

where  $T_f^i$  is the  $i$ -fold iterate of  $T_f$  and each  $g_i$  acts by left scalar multiplication. The same definitions make sense over any division ring  $K' \supseteq K$  to which  $\sigma$  and  $\delta$  have been extended, and then  $T_f$  on  $K'^n$  restricts to  $T_f$  on  $K^n$ .

**Lemma 2.1.** *Under the identification  $R/Rf = K^n$ , left multiplication by  $g \in R$  on  $R/Rf$  is the map  $g(T_f)$ . In particular, for  $b = \sum_{i < n} b_i t^i$  and  $c = \sum_{i < n} c_i t^i$ ,*

$$gb \equiv c \pmod{Rf} \iff g(T_f)(b_0, \dots, b_{n-1}) = (c_0, \dots, c_{n-1}).$$

*Proof.* Left multiplication by  $a \in K$  is scalar multiplication by  $a$ . Left multiplication by  $t$  sends  $\sum_{i < n} b_i t^i$  to  $\sum_{i < n} \sigma(b_i) t^{i+1} + \sum_{i < n} \delta(b_i) t^i$ , and  $t^n \equiv -\sum_{j < n} f_j t^j \pmod{Rf}$ ; so its coefficient vector is  $\sigma(b)C_f + \delta(b) = T_f(b)$ , where  $b = (b_0, \dots, b_{n-1})$ . Iterating,  $t^i$  acts by  $T_f^i$ , and summing,  $g$  acts by  $g(T_f)$ .  $\square$

This is the identification of  $R/Rf$  with the pseudo-linear space  $(K^n, T_f)$  used in [JLL09, §4].

**Proposition 2.2.** *Suppose that for all monic  $f, g \in R$  of degree  $\geq 1$  the map  $g(T_f): K^n \rightarrow K^n$ , where  $n = \deg f$ , is surjective. Then  $R$  is a simple ring, a left PCI domain and a left V-domain.*

*Proof.*  $R$  is left PCI. A cyclic left  $R$ -module is  $R/Rf$  for some  $f \in R$ . If  $f = 0$  this is  $R$  itself. If  $f \neq 0$  it is finite-dimensional over  $K$ , whereas  $R$  is not, so  $R/Rf$  is a proper cyclic module, and we must show that it is injective. Since  $Rf = Ra^{-1}f$  for  $a \in K^\times$ , we may take  $f$  monic; and since  $R/Rf = 0$  if  $\deg f = 0$ , we may take  $\deg f = n \geq 1$ . Put  $M = R/Rf$ . By Baer's criterion,  $M$  is injective if and only if every homomorphism from a left ideal of  $R$  to  $M$  extends to  $R$ . The left ideals are the  $Rg$ , and for  $g \neq 0$  the module  $Rg$  is free on  $g$ , so a homomorphism  $Rg \rightarrow M$  is the same as an element  $m \in M$  (the image of  $g$ ), and it extends to  $R$  if and only if  $m = gm'$  for some  $m' \in M$ . Thus  $M$  is injective if and only if  $gM = M$  for every  $g \neq 0$ . Write  $g = ag_1$  with  $a \in K^\times$  and  $g_1$  monic; then  $gM = g_1M$ . If  $\deg g_1 = 0$  then  $g_1 = 1$ ; if  $\deg g_1 \geq 1$  then  $g_1M = M$  says, by Lemma 2.1, that  $g_1(T_f)$  is surjective on  $K^n$ , which is the hypothesis.

$R$  is simple. Let  $I \neq 0$  be a two-sided ideal and  $0 \neq f \in I$ . Multiplying by the inverse of the leading coefficient we may take  $f$  monic. If  $\deg f = 0$  then  $f = 1 \in I$ . Otherwise  $f(R/Rf) = R/Rf$  by what has just been proved, so  $fb \equiv 1 \pmod{Rf}$  for some  $b \in R$ , and  $1 \in fR + Rf \subseteq I$ .

$R$  is left V. A simple left  $R$ -module is  $R/L$  for a maximal left ideal  $L$ . The zero ideal is not maximal, since  $Rt$  is a non-zero proper left ideal; so  $L \neq 0$ , and  $R/L$  is a proper cyclic module, hence injective.  $\square$

## 2.2. The right side.

**Proposition 2.3.** *If  $c \in K \setminus \sigma(K)$ , then  $tR \cap ctR = 0$ . In particular, if  $\sigma$  is not surjective then  $R$  is not right Ore.*

*Proof.* Suppose  $tr = ctr'$  with  $r, r' \neq 0$ . Comparing degrees,  $\deg r = \deg r' = p$ , say, with leading coefficients  $r_p$  and  $r'_p$ . The leading coefficient of  $tr$  is  $\sigma(r_p)$  and that of  $ctr'$  is  $c\sigma(r'_p)$ , so  $c = \sigma(r_p)\sigma(r'_p)^{-1} = \sigma(r_p r'_p{}^{-1}) \in \sigma(K)$ , a contradiction.  $\square$

The same computation appears in the proof of [JLL09, Proposition 6.1], where it is used to show that  $R_R$  is not uniform. A right PCI domain is right Ore [CF75, 6.17], so Proposition 2.3 already shows that  $R$  is not right PCI when  $\sigma$  is not surjective. For the failure of the right V-property we use a cardinality argument in the spirit of Lawrence [Law77, Theorem 4]; it also gives the failure of the right PCI-property without appeal to [CF75].

**Lemma 2.4.** *Let  $R$  be a countable domain containing non-zero elements  $a, b$  with  $aR \cap bR = 0$ . Then no non-zero countable right  $R$ -module is injective. Consequently  $R$  is neither a right V-ring nor a right PCI ring.*

*Proof.* The right ideals  $b^i aR$  ( $i \in \mathbb{N}$ ) are independent. Indeed, if  $\sum_{i=0}^N b^i a r_i = 0$ , then

$$a r_0 = -b \sum_{i=1}^N b^{i-1} a r_i \in aR \cap bR = 0,$$

so  $r_0 = 0$  because  $R$  is a domain; cancelling  $b$  on the left and repeating gives  $r_i = 0$  for all  $i$ . Each  $b^i aR$  is isomorphic to  $R_R$ , so  $I = \bigoplus_{i \in \mathbb{N}} b^i aR$  is a free right ideal of countably infinite rank. If  $E$  is an injective right  $R$ -module, the restriction map  $\text{Hom}_R(R, E) \rightarrow \text{Hom}_R(I, E)$  is surjective; as  $\text{Hom}_R(R, E) \cong E$  and  $\text{Hom}_R(I, E) \cong E^{\mathbb{N}}$ , this gives  $|E^{\mathbb{N}}| \leq |E|$ , which is impossible if  $E$  is non-zero and countable, since then  $|E^{\mathbb{N}}| = 2^{\aleph_0}$ .

A simple right  $R$ -module  $R/M$  is countable, hence not injective; so  $R$  is not a right V-ring. Moreover  $R/M$  is a proper cyclic module:  $R_R$  is not simple, since  $aR$  is a non-zero right ideal and  $aR \neq R$  (otherwise  $b \in aR \cap bR$ ), so  $R/M \not\cong R_R$ . Hence  $R$  is not a right PCI ring either.  $\square$

**2.3. Reduction.** The Main Theorem is now reduced to the following statement, which is proved in Section 4.

**Theorem 2.5.** *Let  $k$  be a countable commutative field. There are a countable division ring  $K \supseteq k$ , a non-surjective endomorphism  $\sigma$  of  $K$  and a  $\sigma$ -derivation  $\delta$  of  $K$  such that, in  $R = K[t; \sigma, \delta]$ , for all monic  $f, g \in R$  of degree  $\geq 1$  the map  $g(T_f): K^n \rightarrow K^n$ ,  $n = \deg f$ , is surjective.*

*Proof of the Main Theorem from Theorem 2.5.* Let  $K$ ,  $\sigma$  and  $\delta$  be as in Theorem 2.5, and let  $R = K[t; \sigma, \delta]$ . Then  $R$  is a countable principal left ideal domain, and by Proposition 2.2 it is simple, left PCI and left V. Choose  $c \in K \setminus \sigma(K)$ . By Proposition 2.3 we have  $tR \cap ctR = 0$ , so  $R$  is not right Ore; and by Lemma 2.4, applied with  $a = t$  and  $b = ct$ , it is neither a right V-ring nor a right PCI ring. The statements about  $R^{\text{op}}$  are the mirror images, using that a principal left ideal domain is left Noetherian and left hereditary.  $\square$

**Corollary 2.6.** *Let  $R$  be the ring of the Main Theorem.*

- (1) *A right PCI ring need not be left PCI ([Fai73], [CF75, §7, (10)], [Dam79]):  $R^{\text{op}}$ .*
- (2) *A right PCI domain, and a right V-domain, need not be left Ore ([CF75, §7, (7)]):  $R^{\text{op}}$ .*
- (3) *A left V-domain need not be a right V-domain ([CF75, §7, (10)], [JLL09, §1], [BDV18, Remark 2.8]):  $R$ .*
- (4) *There is a left V-domain of the form  $K[t; \sigma, \delta]$  with  $\sigma$  not onto ([JLL09, Remark 6.3]):  $R$ .*

### 3. FREE FIELDS

We recall the little that we need from Cohn's theory of universal fields of fractions [CohSF, Chapter 4 and §5.4], [CohFIR, Chapter 7]. A square matrix  $A$  of size  $p$  over a ring  $S$  is *full* if it cannot be written as  $A = PQ$  with  $P$  of size  $p \times r$ ,  $Q$  of size  $r \times p$  and  $r < p$ . Over a domain, a  $1 \times 1$  matrix ( $a$ ) is full if and only if  $a \neq 0$ . A ring homomorphism  $\phi: S \rightarrow B$  is *fully inverting* if for every full matrix  $A$  over  $S$  the matrix  $\phi(A)$  is invertible over  $B$ .

Fix a set  $Y$  and let  $S = k\langle Y \rangle$  be the free associative  $k$ -algebra on  $Y$ . We use one theorem of Cohn: the free algebra  $S$  is a fir [CohFIR, Corollary 2.5.2], and the universal localization of a fir (or semifir) at the set of all its full matrices is a division ring [CohSF, Theorem 4.5.8, Corollary 4.5.9], [CohFIR, Theorem 7.5.13]. Together with the defining property of a universal localization [CohSF, §4.1], this says the following.

**Theorem 3.1** (Cohn). *There are a division ring  $\mathcal{F}(Y)$  and a fully inverting homomorphism  $\lambda: S \rightarrow \mathcal{F}(Y)$  such that every fully inverting homomorphism  $\phi: S \rightarrow B$  into an arbitrary ring  $B$  factors uniquely as  $\phi = \phi' \lambda$  with  $\phi': \mathcal{F}(Y) \rightarrow B$  a ring homomorphism.*

The division ring  $\mathcal{F}(Y)$  is the *universal field of fractions* of  $S$ ; we call it the *free field* on  $Y$  over  $k$ . The map  $\lambda$  is injective, since  $(a)$  is full for  $a \neq 0$ , and we regard  $S$  as a subring of  $\mathcal{F}(Y)$ , writing  $y$  for  $\lambda(y)$  when no confusion can arise. The universal property has the following consequence, which does most of the work in this section.

**Lemma 3.2** (rigidity). *Let  $\phi, \psi: \mathcal{F}(Y) \rightarrow B$  be ring homomorphisms into a ring  $B$ . If  $\phi\lambda = \psi\lambda$ , then  $\phi = \psi$ .*

*Proof.* For a full matrix  $A$  over  $S$ , the matrix  $\lambda(A)$  is invertible over  $\mathcal{F}(Y)$ , so  $\phi\lambda(A)$  is invertible over  $B$ . Thus  $\phi\lambda$  is fully inverting, and  $\phi$  and  $\psi$  are both factorizations of it through  $\lambda$ ; they coincide by the uniqueness in Theorem 3.1.  $\square$

For instance,  $k$  is central in  $\mathcal{F}(Y)$ : for  $c \in k^\times$ , conjugation by  $c$  is an automorphism of  $\mathcal{F}(Y)$  which agrees with the identity on  $S$ , hence is the identity. So  $\mathcal{F}(Y)$  is a  $k$ -algebra, and every  $k$ -algebra homomorphism out of  $S$  is determined by its values on  $Y$ .

**Lemma 3.3.**  *$\mathcal{F}(Y)$  is generated by  $Y$  as a division ring. In particular, if  $k$  and  $Y$  are countable, so is  $\mathcal{F}(Y)$ .*

*Proof.* Let  $D \subseteq \mathcal{F}(Y)$  be the division subring generated by  $S$ . A square matrix over  $D$  which is invertible over  $\mathcal{F}(Y)$  is invertible over  $D$ : its rows are left linearly independent over  $\mathcal{F}(Y)$ , hence over  $D$ , and a square matrix over a division ring with linearly independent rows is invertible. Hence the corestriction  $S \rightarrow D$  of  $\lambda$  is fully inverting, and it factors as  $\psi\lambda$  with  $\psi: \mathcal{F}(Y) \rightarrow D$ . The composite of  $\psi$  with the inclusion  $D \subseteq \mathcal{F}(Y)$  agrees with the identity on  $S$ , so it is the identity by Lemma 3.2, and  $D = \mathcal{F}(Y)$ . If  $k$  and  $Y$  are countable then so is  $S$ , and the division subring generated by a countable set is countable.  $\square$

Lemma 3.3 is used only to control the cardinality of  $\mathcal{F}(Y)$ .

**Lemma 3.4.** *Let  $\phi: S \rightarrow S$  be a ring endomorphism with a left inverse, that is, a ring endomorphism  $\rho$  of  $S$  with  $\rho\phi = \text{id}_S$ . Then  $\phi$  maps full matrices to full matrices, and  $\lambda\phi: S \rightarrow \mathcal{F}(Y)$  is fully inverting.*

*Proof.* If  $A$  is full of size  $p$  and  $\phi(A) = PQ$  with  $P$  of size  $p \times r$ ,  $Q$  of size  $r \times p$  and  $r < p$ , then  $A = \rho\phi(A) = \rho(P)\rho(Q)$ , contradicting the fullness of  $A$ . So  $\phi(A)$  is full, and  $\lambda\phi(A)$  is invertible over  $\mathcal{F}(Y)$ . (This is [CohSF, Proposition 4.5.1].)  $\square$

**Proposition 3.5** (endomorphisms and derivations prescribed on generators). *Let  $s: Y \rightarrow Y$  be an injective map and  $d: Y \rightarrow \mathcal{F}(Y)$  an arbitrary map. There are an endomorphism  $\sigma$  of  $\mathcal{F}(Y)$  and a  $\sigma$ -derivation  $\delta$  of  $\mathcal{F}(Y)$  with*

$$\sigma(y) = s(y), \quad \delta(y) = d(y) \quad (y \in Y).$$

*Moreover, if  $\hat{s}$  denotes the  $k$ -algebra endomorphism of  $S$  induced by  $s$ , then  $\sigma\lambda = \lambda\hat{s}$ .*

*Proof.* Write  $U = \mathcal{F}(Y)$ .

*Step 1:  $\lambda\hat{s}$  is fully inverting.* Since  $s$  is injective, there is a  $k$ -algebra endomorphism  $\rho$  of  $S$  with  $\rho(s(y)) = y$  for  $y \in Y$  and  $\rho(y') = 0$  for  $y' \in Y \setminus s(Y)$ . Then  $\rho\hat{s}$  is the identity on  $Y$ , hence on  $S$ , and Lemma 3.4 applies.

*Step 2: a homomorphism into triangular matrices.* Let  $\text{UT}_2(U)$  be the ring of upper triangular  $2 \times 2$  matrices over  $U$ ; it is a  $k$ -algebra since  $k$  is central in  $U$ , and the two diagonal entries  $\pi_1, \pi_2: \text{UT}_2(U) \rightarrow U$  are ring homomorphisms. Let  $\theta: S \rightarrow \text{UT}_2(U)$  be the  $k$ -algebra homomorphism with

$$\theta(y) = \begin{pmatrix} s(y) & d(y) \\ 0 & y \end{pmatrix} \quad (y \in Y).$$

Then  $\pi_1\theta = \lambda\hat{s}$  and  $\pi_2\theta = \lambda$ , since both sides are  $k$ -algebra homomorphisms  $S \rightarrow U$  that agree on  $Y$ .

*Step 3:  $\theta$  is fully inverting.* Let  $A = (a_{ij})$  be a full  $p \times p$  matrix over  $S$ . The matrix  $\theta(A) \in M_p(\text{UT}_2(U))$  has entries  $\theta(a_{ij}) = \begin{pmatrix} \hat{s}(a_{ij}) & q_{ij} \\ 0 & a_{ij} \end{pmatrix}$  for certain  $q_{ij} \in U$ . Regard it as a  $2p \times 2p$  matrix over  $U$  whose rows and columns are indexed by pairs  $(i, \epsilon)$  with  $1 \leq i \leq p$  and  $\epsilon \in \{1, 2\}$ , and reorder the rows and columns so that those with  $\epsilon = 1$  come first. This reordering is an isomorphism of  $M_p(M_2(U))$  onto  $M_{2p}(U)$  which maps  $M_p(\text{UT}_2(U))$  onto the ring of block upper triangular matrices with  $p \times p$  blocks, and it sends  $\theta(A)$  to

$$\begin{pmatrix} \lambda\hat{s}(A) & Q \\ 0 & \lambda(A) \end{pmatrix}, \quad Q = (q_{ij}).$$

The diagonal blocks are invertible over  $U$ :  $\lambda(A)$  because  $\lambda$  is fully inverting, and  $\lambda\hat{s}(A)$  by Step 1. Hence this block matrix is invertible, with inverse

$$\begin{pmatrix} \lambda\hat{s}(A)^{-1} & -\lambda\hat{s}(A)^{-1}Q\lambda(A)^{-1} \\ 0 & \lambda(A)^{-1} \end{pmatrix},$$

which is again block upper triangular. So  $\theta(A)$  is invertible in  $M_p(\text{UT}_2(U))$ , and  $\theta$  is fully inverting. By Theorem 3.1,  $\theta = \Theta\lambda$  for a ring homomorphism  $\Theta: U \rightarrow \text{UT}_2(U)$ .

*Step 4: reading off  $\sigma$  and  $\delta$ .* The homomorphism  $\pi_2\Theta: U \rightarrow U$  satisfies  $\pi_2\Theta\lambda = \pi_2\theta = \lambda$ , so  $\pi_2\Theta = \text{id}_U$  by Lemma 3.2. We may therefore write

$$\Theta(u) = \begin{pmatrix} \sigma(u) & \delta(u) \\ 0 & u \end{pmatrix} \quad (u \in U)$$

for maps  $\sigma, \delta: U \rightarrow U$ . That  $\Theta$  is additive says that  $\sigma$  and  $\delta$  are additive; that  $\Theta(1)$  is the identity matrix says that  $\sigma(1) = 1$  and  $\delta(1) = 0$ ; and comparing the entries of  $\Theta(uv) = \Theta(u)\Theta(v)$  gives

$$\sigma(uv) = \sigma(u)\sigma(v), \quad \delta(uv) = \sigma(u)\delta(v) + \delta(u)v.$$

So  $\sigma$  is an endomorphism of  $U$  and  $\delta$  is a  $\sigma$ -derivation (this is the standard description of twisted derivations by triangular matrices, [CohSF, §2.1]). Finally,  $\Theta\lambda = \theta$  gives  $\sigma(y) = s(y)$  and  $\delta(y) = d(y)$  for  $y \in Y$ , and  $\sigma\lambda = \pi_1\Theta\lambda = \pi_1\theta = \lambda\hat{s}$ .  $\square$

The argument is that of [CohFIR, Theorem 7.5.17], which extends ordinary derivations to universal fields of fractions, with a twist by  $\sigma$  inserted.

**Lemma 3.6** (non-surjectivity). *In the situation of Proposition 3.5, let  $y_0 \in Y \setminus s(Y)$ . Then  $y_0 \notin \sigma(\mathcal{F}(Y))$ . In particular  $\sigma$  is not surjective.*

*Proof.* Write  $U = \mathcal{F}(Y)$ . Let  $\hat{\tau}$  be the  $k$ -algebra automorphism of  $S$  with  $\hat{\tau}(y_0) = y_0 + 1$  and  $\hat{\tau}(y) = y$  for  $y \neq y_0$ . Its inverse is a left inverse, so  $\lambda\hat{\tau}$  is fully inverting by Lemma 3.4, and it

factors as  $\tau\lambda$  for an endomorphism  $\tau$  of  $U$ . Since  $y_0 \notin s(Y)$ , the endomorphism  $\hat{\tau}$  fixes every element of  $s(Y)$ , so  $\hat{\tau}\hat{s} = \hat{s}$ . Hence

$$\tau\sigma\lambda = \tau\lambda\hat{s} = \lambda\hat{\tau}\hat{s} = \lambda\hat{s} = \sigma\lambda,$$

and  $\tau\sigma = \sigma$  by Lemma 3.2. Thus  $\tau$  fixes  $\sigma(U)$  pointwise, whereas  $\tau(y_0) = y_0 + 1 \neq y_0$ . So  $y_0 \notin \sigma(U)$ .  $\square$

*Remark 3.7.* Cohn uses precisely this set-up in [Coh77, §5] and [CohSF, §5.9]: a free algebra on generators indexed by  $\Lambda \times \mathbb{N}^2$ , an endomorphism shifting one index (which has a left inverse and therefore preserves full matrices) and a derivation shifting the other, both extended to the free field. In [Coh77] the non-surjectivity of the extended endomorphism is stated to be clear; Lemma 3.6 supplies the argument.

#### 4. THE CONSTRUCTION

**Construction 4.1.** Let  $k$  be a countable commutative field. Let  $Z$  be a countably infinite set, let

$$Y = \{y_i : i \in \mathbb{N}\} \sqcup Z,$$

and let  $K = \mathcal{F}(Y)$  be the free field on  $Y$  over  $k$ ; it is countable by Lemma 3.3. Let  $s : Y \rightarrow Y$  be the map with  $s(y_i) = y_{i+1}$  and  $s(z) = z$  for  $z \in Z$ . It is injective, and  $y_0 \notin s(Y)$ .

A *task* is a triple  $\tau = (f, g, e)$  with

$$f = (f_0, \dots, f_{n-1}) \in K^n, \quad g = (g_0, \dots, g_{m-1}) \in K^m, \quad e \in K^n,$$

where  $n, m \geq 1$ . We think of  $f$  and  $g$  as the coefficient lists of the monic polynomials  $t^n + \sum_{i < n} f_i t^i$  and  $t^m + \sum_{j < m} g_j t^j$ , and of  $e$  as a right-hand side; but note that a task is defined in terms of  $K$  alone, before  $\sigma$  and  $\delta$  have been chosen. Likewise the companion matrix  $C_f \in M_n(K)$  depends only on the list  $f$ . The set  $\mathcal{T}$  of all tasks is countable, since  $K$  is.

Because  $\mathcal{T}$  is countable and  $Z$  is infinite, we may choose, for each task  $\tau = (f, g, e)$  with  $f \in K^n$  and  $g \in K^m$ , a subset

$$X_\tau = \{x_{l,j}^\tau : 1 \leq l \leq n, 0 \leq j \leq m-1\} \subseteq Z$$

of  $nm$  elements, in such a way that the sets  $X_\tau$  ( $\tau \in \mathcal{T}$ ) are pairwise disjoint. For each  $\tau$  and  $0 \leq j \leq m-1$  let  $x_j^\tau = (x_{1,j}^\tau, \dots, x_{n,j}^\tau) \in K^n$  be the vector of generators in the  $j$ th column of  $X_\tau$ .

Define  $d : Y \rightarrow K$  as follows. On  $\{y_i\}$  and on  $Z \setminus \bigcup_\tau X_\tau$  let  $d = 0$ . On  $X_\tau$ , for  $\tau = (f, g, e)$ , writing  $x_j = x_j^\tau$  and  $C = C_f$ , let  $d$  be given in vector form by

$$(1) \quad d(x_j) = x_{j+1} - x_j C \quad (0 \leq j \leq m-2), \quad d(x_{m-1}) = e - \sum_{j=0}^{m-1} g_j x_j - x_{m-1} C;$$

that is,  $d(x_{l,j}^\tau)$  is the  $l$ th coordinate of the right-hand side. These are elements of  $K$ , and since the sets  $X_\tau$  are disjoint,  $d$  is well defined.

Finally, let  $(\sigma, \delta)$  be an endomorphism and a  $\sigma$ -derivation of  $K$  with  $\sigma(y) = s(y)$  and  $\delta(y) = d(y)$  for  $y \in Y$ , as provided by Proposition 3.5, and let  $R = K[t; \sigma, \delta]$ .

*Proof of Theorem 2.5.* We show that  $K$ ,  $\sigma$ ,  $\delta$  of Construction 4.1 have the required properties.  $K$  is countable, and  $\sigma$  is not surjective by Lemma 3.6, applied to  $y_0 \notin s(Y)$ .

Now let  $f, g \in R$  be monic of degrees  $n, m \geq 1$  and let  $e \in K^n$ ; we must find  $v \in K^n$  with  $g(T_f)(v) = e$ . Let  $\tau = (f, g, e) \in \mathcal{T}$  be the task formed by the coefficient lists of  $f$  and  $g$  and by

$e$ , and write  $x_j = x_j^\tau$  and  $C = C_f$ . The coordinates of each  $x_j$  lie in  $Z$ , so they are fixed by  $\sigma$ , and  $\delta$  acts on them by  $d$ . Hence

$$T_f(x_j) = \sigma(x_j)C + \delta(x_j) = x_jC + d(x_j) \quad (0 \leq j \leq m-1),$$

and by (1),

$$T_f(x_j) = x_{j+1} \quad (0 \leq j \leq m-2), \quad T_f(x_{m-1}) = e - \sum_{j=0}^{m-1} g_j x_j.$$

By induction,  $T_f^j(x_0) = x_j$  for  $0 \leq j \leq m-1$ , and then  $T_f^m(x_0) = T_f(x_{m-1}) = e - \sum_{j < m} g_j x_j$ . Since  $g$  is monic of degree  $m$ ,

$$g(T_f)(x_0) = \sum_{j=0}^{m-1} g_j T_f^j(x_0) + T_f^m(x_0) = \sum_{j=0}^{m-1} g_j x_j + \left( e - \sum_{j=0}^{m-1} g_j x_j \right) = e.$$

So  $v = x_0$  solves the equation. □

This completes the proof of the Main Theorem.

## 5. REMARKS

*Remark 5.1* (on the shape of the construction). Two features of the construction are forced.

- (1) The derivation cannot be omitted or made inner. If  $\delta = 0$ , then  $tR \subseteq Rt$  and  $Rt$  is a proper two-sided ideal of  $K[t; \sigma]$ , so the ring is not simple. If  $\delta$  is inner,  $\delta(a) = ca - \sigma(a)c$  for some  $c \in K$ , then the substitution  $t' = t - c$  gives  $K[t; \sigma, \delta] = K[t'; \sigma]$ . In particular  $K$  cannot be commutative: if  $K$  is commutative and  $\sigma \neq \text{id}$ , every  $\sigma$ -derivation is inner, because for  $b$  with  $\sigma(b) \neq b$  the identity  $\delta(ab) = \delta(ba)$  yields  $\delta(a) = c(a - \sigma(a))$  with  $c = \delta(b)(b - \sigma(b))^{-1}$  [Coh77, §2].
- (2) The solutions of the equations  $g(T_f)(v) = e$  have to be adjoined as free generators. What makes the construction work is that on the free field the values of  $\sigma$  and  $\delta$  on the generators are subject to no constraints whatsoever (Proposition 3.5); an Ore extension or a crossed product of  $K$  would impose commutation relations between the new elements and  $K$  which  $\delta$  would have to respect.

By contrast, the choice to fix the new generators under  $\sigma$ , and to prescribe  $T_f$  rather than  $\delta$  on them, is a matter of convenience, but a decisive one. If one prescribes  $\delta(x_j) = x_{j+1}$  directly, one finds  $T_f^i(x_0) = \sum_{j \leq i} x_j N_{i,j}$  for matrices  $N_{i,j}$  built from  $\sigma$  and  $\delta$  applied to the coefficients of  $f$ , so the required value of  $\delta$  on the last generator depends on  $\sigma$  and  $\delta$ , and one is led to build  $K$  as the union of a tower of free fields, each solving the equations of the previous one, with the attendant compatibility checks. Prescribing  $T_f(x_j)$  makes the required values of  $d$  depend only on the coefficients of  $f$ ,  $g$  and  $e$ , so that a single free field suffices.

*Remark 5.2* (uncountable  $k$ ). For an arbitrary commutative field  $k$ , take  $|Z| = \max(|k|, \aleph_0)$  in Construction 4.1. Then  $|K| = |Z|$  by Lemma 3.3, so  $|\mathcal{T}| = |Z|$  and the sets  $X_\tau$  can still be chosen, and the proof of Theorem 2.5 goes through unchanged. One obtains a left PCI domain  $K[t; \sigma, \delta]$  which is not right Ore, hence not right PCI by [CF75, 6.17]. Only the failure of the right V-property was derived from countability.

*Remark 5.3* (on the literature used). Apart from Baer's criterion and the elementary theory of Ore extensions, the proof uses from the literature only Theorem 3.1: the free algebra has a universal field of fractions with the universal property of a localization. Lemmas 3.2–3.4, Proposition 3.5 and Lemma 3.6 are derived from it. In particular, Cohn's theorem that honest homomorphisms between semifirs extend to their universal fields of fractions [CohSF, Theorem 4.5.10] is not needed: the extensions used here are obtained directly from the universal

property, and the identifications between free fields on different sets of generators that a tower would require do not arise.

*Remark 5.4* (further properties of  $R$ ). By [JLL09, Theorem 6.2], the maximal right quotient ring of the left V-domain  $R$  is directly infinite, and by [JLL09, Proposition 6.1] no cyclic right module  $R/fR$  with  $f \neq 0$  is injective and every principal right ideal of  $R$  is closed. The proof of Lemma 2.4 shows more than was used: every non-zero injective right  $R$ -module  $E$  satisfies  $|E|^{\aleph_0} \leq |E|$ , hence  $|E| \geq 2^{\aleph_0}$ .

## REFERENCES

- [BDV18] M. Behboodi, A. Daneshvar, M. R. Vedadi, *Several generalizations of the Wedderburn–Artin theorem with applications*, Algebr. Represent. Theory **21** (2018), 1333–1342.
- [Boy74] A. K. Boyle, *Hereditary QI-rings*, Trans. Amer. Math. Soc. **192** (1974), 115–120.
- [BG75] A. K. Boyle, K. R. Goodearl, *Rings over which certain modules are injective*, Pacific J. Math. **58** (1975), 43–53.
- [Coh77] P. M. Cohn, *A construction of simple principal right ideal domains*, Proc. Amer. Math. Soc. **66** (1977), 217–222.
- [CohSF] P. M. Cohn, *Skew Fields: Theory of General Division Rings*, Encyclopedia of Mathematics and its Applications **57**, Cambridge University Press, 1995.
- [CohFIR] P. M. Cohn, *Free Ideal Rings and Localization in General Rings*, New Mathematical Monographs **3**, Cambridge University Press, 2006.
- [Coz70] J. H. Cozzens, *Homological properties of the ring of differential polynomials*, Bull. Amer. Math. Soc. **76** (1970), 75–79.
- [CF75] J. H. Cozzens, C. Faith, *Simple Noetherian Rings*, Cambridge Tracts in Mathematics **69**, Cambridge University Press, 1975.
- [Dam79] R. F. Damiano, *A right PCI ring is right Noetherian*, Proc. Amer. Math. Soc. **77** (1979), 11–14.
- [Fai73] C. Faith, *When are proper cyclics injective?*, Pacific J. Math. **45** (1973), 97–112.
- [JLL09] S. K. Jain, T. Y. Lam, A. Leroy, *Ore extensions and V-domains*, in: Rings, Modules and Representations, Contemp. Math. **480**, Amer. Math. Soc., 2009, 249–262.
- [JS09] S. K. Jain, A. K. Srivastava, *Rings determined by the properties of cyclic modules and duals: a survey-I*, preprint, 2009, <https://leroy.perso.math.cnrs.fr/Talks/Jain.pdf>.
- [Law77] J. Lawrence, *A countable self-injective ring is quasi-Frobenius*, Proc. Amer. Math. Soc. **65** (1977), 217–220.